

Single Organ Segmentation (Model=SegFormer)

(Intestinal Veins, Ureter, Inferior mesenteric artery (IMA), Vesicular glands (Vglands), Pancreas)

(Framework=PyTorch, Encoder=resnet50, Treshold=0.40, trainig size=0.8, test_size=0.1, vaidation size=0.1)

a) **Intestinal Veins Data** (Train image=853, Test image =106, validation image=108)

image size 128x128 (Model: SegFormer)

Number of Epochs	10 Epoch	30 Epoch	50 Epoch
IoU	0.4962	0.5201	0.5692
Precision	0.6039	0.6092	0.6704
Recall	0.7333	0.7715	0.7546
Specificity	0.9949	0.9957	0.9971
F1 Skore	0.6345	0.6612	0.6945
Running Time (seconds)	375.44	1166.39	1770.61

image size 256x256 (Model: SegFormer)

Number of Epochs	10 Epoch	30 Epoch	50 Epoch
IoU	0.5265	0.5597	0.6500
Precision	0.5696	0.6055	0.7162
Recall	0.8620	0.8692	0.8684
Specificity	0.9939	0.9944	0.9965
F1 Skore	0.6710	0.6971	0.7736
Running Time (seconds)	407.36	1232.22	1771.55

b) **Ureter Data** (Train image=1020, Test image=128, validation image=127)

image size 128x128 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.4280	0.4720	0.5485
Precision	0.6180	0.6963	0.6123
Recall	0.5650	0.5856	0.8048
Specificity	0.9962	0.9974	0.9950
F1 Skore	0.5467	0.5916	0.6782
Running Time (seconds)	392.82	973.48	2046.38

image size 256x256 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.4630	0.5455	0.5704
Precision	0.5214	0.7664	0.6368
Recall	0.7868	0.6513	0.7946
Specificity	0.9926	0.9981	0.9959
F1 Skore	0.5973	0.6676	0.6926
Running Time (seconds)	427.35	1168.73	2012.62

c) **Inferior mesenteric artery data** (Train image=872, Test image=109, validation image=109)

image size 128x128 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.5321	0.5992	0.6207
Precision	0.5843	0.7388	0.7151
Recall	0.8460	0.7541	0.7960
Specificity	0.9859	0.9947	0.9932
F1 Skore	0.6737	0.7229	0.7425
Running Time (seconds)	379.47	863.67	1823.74

image size 256x256 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.5552	0.6201	0.6410
Precision	0.6287	0.7545	0.7264
Recall	0.8382	0.7397	0.8378
Specificity	0.9882	0.9953	0.9926
F1 Skore	0.7005	0.7342	0.7605
Running Time (seconds)	408.85	1098.11	1825.51

d) **Vglands Data** (Train image=958, Test image=121, validation image=119)

image size 128x128 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.4117	0.4621	0.5392
Precision	0.6396	0.6304	0.6576
Recall	0.5241	0.5775	0.7061
Specificity	0.9933	0.9948	0.9920
F1 Skore	0.5259	0.5740	0.6472
Running Time (seconds)	375.13	1206.53	1912.36

image size 256x256 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.5121	0.5557	0.5949
Precision	0.6665	0.6588	0.7171
Recall	0.6275	0.7392	0.7565
Specificity	0.9949	0.9936	0.9951
F1 Skore	0.6194	0.6746	0.7167
Running Time (seconds)	408.73	1354.31	1973.80

e) **Pancreas Data** (Train image=938, Test image=117, validation image=118)

image size 128x128 (Model: SegFormer)

Number of Epochs	10	30	50 (46 e.s.)
IoU	0.4942	0.5122	0.5797
Precision	0.6805	0.6922	0.6828
Recall	0.6200	0.6293	0.7783
Specificity	0.9941	0.9956	0.9938
F1 Skore	0.6104	0.6293	0.7035
Running Time (seconds)	370.31	897.63	1736.12

image size 256x256 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.5278	0.5494	0.5901
Precision	0.6193	0.7118	0.7297
Recall	0.7317	0.6824	0.7451
Specificity	0.9903	0.9954	0.9937
F1 Skore	0.6504	0.6723	0.7140
Running Time (seconds)	403.63	948.24	1958.42

image size 512x512 (Model: SegFormer)

Number of Epochs	10	30	50
IoU	0.5516	0.6398	0.6831
Precision	0.6508	0.7183	0.7725
Recall	0.7749	0.8104	0.8559
Specificity	0.9895	0.9940	0.9942
F1 Skore	0.6813	0.7473	0.7997
Running Time (seconds)	1080.54	3106.61	5849.35